

Paper:

TSV Manufacturing Fault Modeling and Diagnosis Based on Multi-Tone Dither

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The faults in through-silicon via (TSV) have a critical impact on the reliability and yield of a three-dimensional integrated circuit (3-D IC). With the significant increase in the number of TSVs used in 3-D IC, the testing of TSVs for manufacturing faults poses certain serious challenges especially weak fault testing, and therefore it is important to have effective Design-For-Test (DFT) techniques. In this paper, we present a method for TSV testing using multi-tone dither signal, based on electrical characteristic analysis. This method mainly observes the differences in the root mean square (RMS) value of the output signal voltage between faultless and faulty TSV circuits to detect manufacturing faults, and uses only passive components such as metal lines, without consuming additional power for the testing. With regard to the common manufacturing faults such as voids and pinholes, the electrical characteristics of faulty TSVs are modeled and analyzed, and analytic equations of the faults, which are based on characteristic parameters, are explored. The ground-signal-TSV (GS-TSV) equivalent electrical model with manufacturing faults is simulated and tested by using a multi-tone dither test signal, which is generated by modulating an RF signal with an optimized multi-tone signal. The peak-to-average ratio (PAR) is used as the test evaluation parameter to determine the type and size of the fault. The simulation results demonstrate the effectiveness of the multi-tone dither test method in the detection of voids (as low as ohm level) and pinholes (up to mega ohm level). It is obvious that this method performs better in the diagnosis of weak manufacturing faults in TSVs.

Keywords: fault diagnosis, multi-tone dither test, TSV, electrical model of manufacturing fault, multi-tone signal

1. Introduction

TSV-based 3-D ICs have significant advantages in terms of power consumption, performance, form factor, heterogeneous integration, etc. [1]. With the development of 3-D IC system design goals allowing high-speed data transmission between chips up to the range of TB/s, the number of input/output (I/O) must be maximized. This directly affects the physical size and density of TSV interconnects; that is, the diameter and pitch of TSVs is minimized [2, 3]. However, TSVs with small diameter and small pitch are susceptible to mechanical and thermal stress. In addition, the complexity and immaturity of the fabrication process inevitably cause some manufacturing faults in TSVs; for example, the formation of voids due to the failure to completely fill out the etched spaces or thermal effects, creation of pinholes in the insulating layer because of the differences in the thermal expansion coefficients of the dielectric and the substrate, formation of cracks due to thermal and mechanical loadings during the TSV formation process, etc. [4, 5]. In addition, there may be weak faults, which are small size faults of the above manufacturing faults, including microvoids or underfillings (both leading to weak opens) and pinholes in the oxide (causing weak short faults). A weak fault does not directly affect the TSV function, but causes a small delay in the circuit. All of these faults degrade the performance of TSV channels in terms of electrical characteristics [2], and their existence will cause attenuation or even complete distortion of the TSV transmission signal [6]. At the same time, in terms of high-frequency and high-speed operation, the signal transmission performance is extremely sensitive to changes in TSV size [7]. The more the amount of TSVs embedded in a 3-D IC, the more the electrical fault ratio, and a single TSV fault can destroy the entire stacked chips, thus affecting the final yield of 3-D ICs. Furthermore, a weak fault with little impact during normal operation may degenerate into a catastrophic fault due to electro-migration and various other reasons. Therefore, TSV fault diagnosis is an important means of



assuring TSV reliability and 3-D IC quality.

At present, many universities and research institutions are conducting researches on TSV fault detection. In particular, the electrical test [8] has attracted considerable attention. It is not only capable of performing the TSV fault test, but is also capable of performing the function and performance test of 3-D ICs. Lou et al. [9] tested the completely open fault of TSVs by applying a square wave signal to a bridge circuit having a large area overhead. Bian et al. [10] proposed an improved CAF-WAS test method, which can accurately predict the range of open and leakage faults and save test time. Deutsch et al. [11, 12] used open TSVs as capacitive loads of their driving gates and measured the variation in propagation delay by using ring oscillators to detect the resistive open and leakage faults. Yu et al. [13] proposed a voltage detection method to detect leakage faults and micro-bump open faults; however, weak faults could not be detected. Basith et al. [14, 15] introduced a non-contact probe with capacitive coupling to test TSV, which avoids the damage caused by traditional probe testing and supports the requirement of high density and small-pitch for TSV detection. Arumí et al. [16] proposed a BIST architecture, which adds a reasonable area overhead in a short-test application time to improve the detection of weak bridging and open defects; however, this can only be used for pre-bonding test.

Although recent studies have improved the effectiveness of TSV testing, there is still a need for the development of methodologies with a better performance in terms of fault coverage, area overhead, and test application time. From the above, it can be seen that most of the existing TSV electrical test methods can only detect one or two kinds of logic circuit faults, and there has been no in-depth study on the shape, size, and position corresponding to the manufacturing fault, especially for the detection of weak faults. In addition, most of the test peripherals have large area overhead, and consumes additional power for the testing. Therefore, it is essential to study the relevant test methods for TSV manufacturing faults.

In this paper, a multi-tone dither [17] test method is proposed to diagnose TSV manufacturing faults. This method is based on the electrical test, and utilizes a multi-tone dither signal based on a multi-tone signal to stimulate the equivalent circuit of faulty TSVs. The output response of the circuit is then observed and analyzed to perform fault diagnosis. The test excitation signals used in the existing TSV fault electrical detection methods are mostly simple single-source signals such as DC signal, sine signal, pulse signal, etc. The main body of the test excitation selected by this method is the multi-tone signal, which contains a variety of frequency, amplitude and phase information. It is a superposition of multiple tones generated by a reference cosine signal through a random phase shift that obeys the Gaussian distribution. This change in phase shift during testing contributes to a significant change in the RMS value of the voltage (V_{RMS}) of the faulty circuit output signal. In order to use this phase shift and change in V_{RMS} effectively, the peak-to-average ratio (PAR) of the circuit under test is used as the

test index [18]. The TSV fault type and size is estimated by analyzing and comparing the degree of difference between the PAR of fault-free and faulty TSVs. Due to the complexity of multi-tone dither signal synthesis, the PAR of the output signal has a greater response to the internal information of the equivalent circuit of the faulty TSVs, which has a higher resolution for the TSV test of weak fault.

This paper is organized as follows. Section 2 introduces and analyzes the electrical models of GS-TSV with manufacturing faults. Section 3 provides a multi-tone dither test method to diagnose TSV manufacturing faults, and discusses the parameters of multi-tone signal selection. Section 4 shows the simulation results of the faulty TSVs models developed in Section 2, which verifies the effectiveness of this method for weak manufacturing fault tests. Finally, Section 5 concludes this study.

2. Electrical Model of TSV

Accurate modeling of the TSV is crucial for its electrical simulation accuracy. It is therefore very important to understand the electrical characteristics of the TSV, which helps in TSV testing and is critical to enhancing the quality of TSV-based 3-D ICs. In this section, a high-frequency extensible electrical model of TSVs is established, which is represented by the RLGC analytic equation based on a lumped model.

2.1. Electrical Model of Fault-Free TSVs

GS-TSV, one of which is the signal TSV and the other is the ground TSV, is chosen as the study object. Considering the silicon substrate effect and the proximity and skin effects of conductors in the high-frequency case, GS-TSV is equivalent to a lumped circuit composed of RLGC. The equivalent circuit model is shown in Fig. 1(a).

The formulas for the calculation of the parameters in Fig. 1(a) are given in [1].

2.2. Electrical Model of TSVs with Fault

TSV, which is the conductive channel connecting the devices in the upper and lower layers, is a key component of 3-D ICs. Its reliability plays a decisive role in the overall performance of a 3-D IC. Due to the limitations of the current TSV technology, there would inevitably be some manufacturing faults during the TSV fabrication and bonding processes, which affect the TSV characteristics and the signal transmission performance of the TSV to varying degrees. The analysis and electrical modeling of the common void and pinhole faults are described below, and the analytical equations of TSV faults based on the fault factors (physical size and material properties) are explored.

2.2.1. Electrical Model of TSVs with Void Fault

Incomplete filling or stress effects of the TSV copper conductor will form voids [19]. The presence of voids

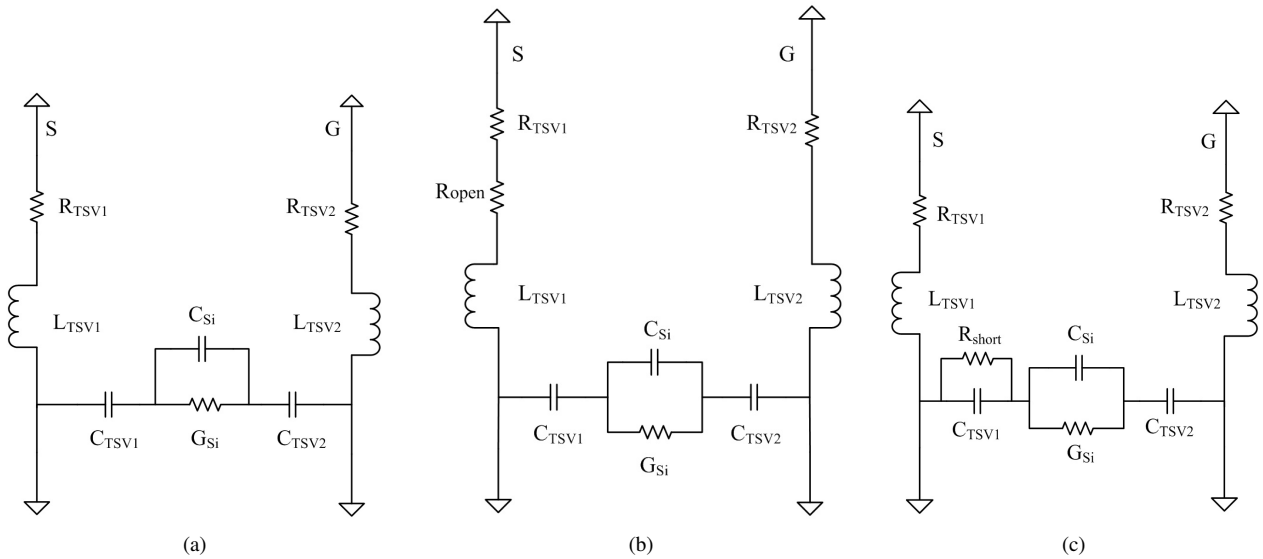


Fig. 1. (a) Equivalent circuit of GS-TSV. (b) Equivalent circuit of GS-TSV with void. (c) Equivalent circuit of GS-TSV with pinhole.

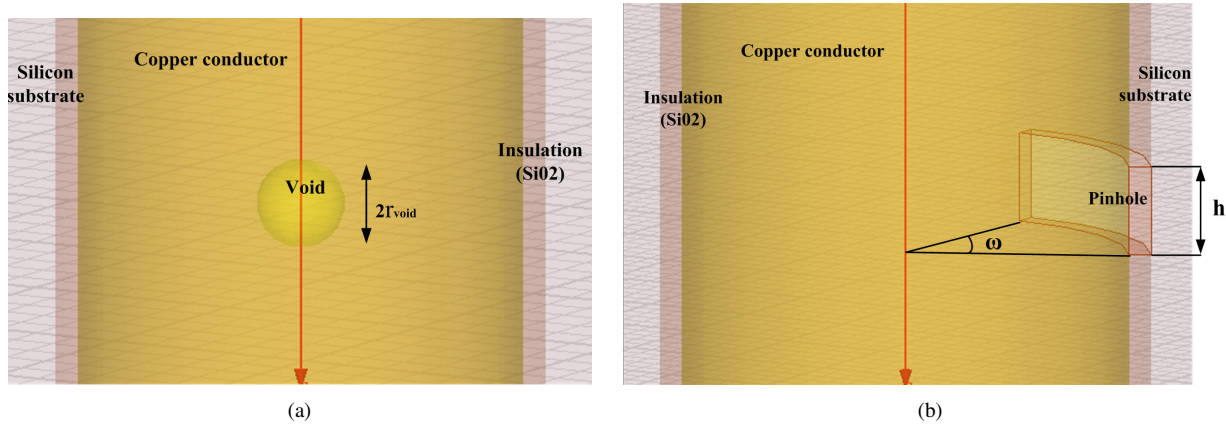


Fig. 2. (a) TSV physical structure model with void. (b) TSV physical structure model with pinhole.

leads to sudden changes in the impedance of the TSV, which also results in partial reflection of the transmitted signals. It also hinders the flow of current and increases the delay time [20]. When a void appears in the TSV center conductor, it creates an open fault. The void reduces the effective for signal transmission and forms a high impedance path [21]. At the same time, due to the appearance of the void, the air dielectric together with the surrounding copper conductor forms a capacitor, resulting in additional power consumption while increasing the transmission signal attenuation, which seriously affects the transmission performance of the TSV. The capacitance of the void is very close to zero and can thus be neglected; therefore, the void is modeled as a resistor R_{open} . When the value of R_{open} equals zero, TSV can be considered to be fault-free; when the value of R_{open} tends to infinity, TSV has a full open fault. With the increase in the size of the void, the value of R_{open} can vary from a few milliohms to infinity. Along with the GS-TSV electrical model, the equivalent circuit model of GS-TSV with a void fault is

show in **Fig. 1(b)**.

When a current flows through a conductor, it always tends to flow along a path with smaller impedance. When there is a void in the TSV, the current will flow through a path without a void, causing an increase in the equivalent resistance of the entire TSV.

The void is modeled by a sphere with radius $r_{void} = kr_{TSV}$, where k denotes the percentage of the radius of the void to the radius of the TSV, as shown in **Fig. 2(a)**.

In order to calculate the resistance (R_{open}) of the void in this structure, the TSV is divided into three cylindrical conductors, one of which includes a spherical void whose height is from the upper point of the sphere to the lower point of the sphere. Therefore, the TSV resistance R'_{TSV1} , which has a void, is divided into three parts, namely the resistance R_2 containing the void and the resistances R_1 and R_3 that do not contain the void; thus, $R'_{TSV1} = R_1 + R_2 + R_3$. Assuming that the void occurs in the middle part of the TSV central conductor, the resistance values of the upper and lower parts are the same,

given by: $R_1 = R_3 = R$. Then the total resistance can be expressed by: $R'_{TSV1} = 2R + R_2$. To model R'_{TSV1} with a nonuniform current distribution at high frequencies, the skin depth (δ) has to be determined to calculate the TSV resistance, which is composed of DC and AC resistances. The resistances R_1 and R_3 are calculated by Eqs. (1)–(3):

$$R = \sqrt{R'^2_{ac-TSV} + R'^2_{dc-TSV}} \quad \dots \quad (1)$$

$$R'_{dc-TSV} = \rho \frac{\frac{h_{TSV}}{2} - r_{void}}{\pi r^2_{TSV}} \quad \dots \quad (2)$$

$$R'_{ac-TSV} = \rho \frac{\frac{h_{TSV}}{2} - r_{void}}{\pi \delta (2r_{TSV} - \delta)} \quad \dots \quad (3)$$

where r_{TSV} , h_{TSV} , r_{void} , δ and ρ are the TSV radius, TSV height, void radius, skin depth and the resistivity of copper, respectively.

The resistance of R_2 is calculated by Eqs. (4)–(8):

$$R_2 = \sqrt{R'^2_{2-ac-TSV} + R'^2_{2-dc-TSV}} \quad \dots \quad (4)$$

Assuming that the radius of cross-section of the void at the fixed height $z = x$ is r , the remaining conductive area S is: $S = \pi(r^2_{TSV} - r^2_{void} + x^2)$. The resistance for a very small height dx is expressed by: $dR'_2 = \rho(dx/S) = \rho dx / (\pi(r^2_{TSV} - r^2_{void} + x^2))$. Therefore the total DC resistance of the TSV containing the entire void can be given by:

$$\begin{aligned} R'_{2-dc-TSV} &= \frac{2\rho}{\pi \sqrt{r^2_{TSV} - r^2_{void}}} \arctan \left(\frac{r_{void}}{\sqrt{r^2_{TSV} - r^2_{void}}} \right) \\ &\dots \dots \dots (5) \end{aligned}$$

In the case of high-frequency, alternating current flows close to the surface of the conductor due to the formation of eddy current, which is called the “skin effect,” and a skin depth (δ) is also formed around the void in the TSV. It is easy to understand that the fault is positioned in the conductor material deeper than the conductor skin depth (δ). In order to calculate the total AC resistance of the TSV, which contains the entire void, the first consideration is whether the surface of the void can form a complete skin depth, which is due to the limited cross-section of the copper conductor. The value of $R'_{2-dc-TSV}$ is determined by the size of the void and the skin depth at the test frequency, which is mainly considered in two cases:

(1) The cross-section length of the remaining Cu conductor is greater than 4δ ;

$$R'_{2-ac-TSV} = \int_{-r_{void}}^{r_{void}} \frac{\rho dx}{\pi (2\delta \sqrt{r^2_{void} - x^2} + \delta^2)} \quad (6)$$

(2) The cross-section length of the remaining Cu conductor is less than 4δ ;

$$R'_{2-ac-TSV} = \rho \frac{2r_{void}}{\pi (r^2_{TSV} - r^2_{void})} \quad \dots \quad (7)$$

In addition, the skin depth is expressed by $\delta = \sqrt{\rho / \pi f \mu}$, where ρ , f , and μ are the resistivity of the metal, the signal frequency, and the permeability of vacuum, respectively.

Therefore, the equivalent resistance of the TSV void is calculated as follows:

$$R_{open} = R'_{TSV1} - R_{TSV1} \quad \dots \quad (8)$$

where R_{TSV1} represents the fault-free TSV resistance.

2.2.2. Electrical Model of TSVs with Pinhole Fault

Pinholes can be formed in the TSV insulation layer due to non-uniform dielectric deposition or stress damage of the insulation layer. Pinholes may also be the result of impurities inside the insulator. Theoretically, the impedance between the TSV and the silicon substrate is infinite; however, when pinholes are created in the insulation layer, an undesirable leakage current can flow between the TSV and the substrate, which results in a low breakdown voltage and a possible resistive short between them. This situation can lead to a low transition or an attenuation of the inter-chip signal, which can even result in failure. A pinhole can be modeled as an impedance R_{short} in parallel with the TSV insulator capacitor C_{TSV} . The magnitude of the R_{short} value represents the degree of resistive short of the TSV; that is, the smaller the value of R_{short} , the more severe the pinhole fault. With the decrease in short impedance R_{short} , the TSV transmission delay and the power consumption of the leakage current will increase [9]. In addition, pinhole defects get aggravated with the aging of circuit, which influences the reliability of the chip. The equivalent circuit model of GS-TSV with pinhole fault can be obtained by combining the GS-TSV equivalent circuit model, as shown in **Fig. 1(c)**.

The influences of pinhole fault locations are also studied by 3D simulation. It is observed that the model parameters are not affected by the physical location.

From the analysis of the pinhole structure, it is seen that the current leakage is caused by insulation layer damage when a high frequency increases the value of C_{TSV} . Usually, the pinhole is filled with air and when a 3-D IC is working, the copper conductor, the pinhole, and the silicon substrate constitute a MOS structure capacitor C_{short} , whose capacitive reactance is equivalent to that of a low impedance path. Therefore, the pinhole fault is equivalent to a capacitive impedance model. The physical structure model of the TSV pinhole is shown in **Fig. 2(b)**, where ω is the radian corresponding to the pinhole and h is the height of the pinhole. A mapping relationship between the physical parameters (ω and h) of the pinhole and the impedance value can be obtained using this model. The parasitic capacitance C_{air} of the insulating layer can be obtained according to the Gauss theorem and the coaxial cable capacitance formula [22], and $C_{air} = 2\pi\epsilon_{air}h / \ln(1 + t_{ox}/r_{TSV1})$, where h and r_{TSV1} are the height and radius of the capacitor, respectively, and t_{ox} is the insulator thickness.

Moreover, C_{air} can be seen as a parallel connection of several MOS capacitors. The ratio of pinhole volume to

the TSV insulation volume can be denoted as $\gamma = \omega/2\pi$, then $C_{short} = \gamma \cdot C_{air}$. Therefore, the equivalent impedance R_{short} of pinhole is expressed as shown below:

$$R_{short} = \frac{\ln\left(1 + \frac{t_{ox}}{r_{TSV1}}\right)}{2\pi f \epsilon_0 \epsilon_{r,air} \omega h} \quad (9)$$

where $\epsilon_{r,air}$ is the relative permittivity of air and f is the signal frequency.

The mathematical extraction equations for the above faults were verified using a 3D electromagnetic field simulator—Ansys HFSS (High Frequency Structure Simulator) and a circuit-level simulator—Keysight ADS (Advanced Design System), for various parameters of the structure. Specifically, the results of the s-parameter from the ADS and HFSS were compared in the frequency-domain, where the faulty GS-TSV equivalent circuit is simulated in the ADS. The simulation results show that the two s-parameter curves are basically consistent in the frequency range of 1–20 GHz, and the relative errors of S_{21} are all less than 3%, which verifies the faulty GS-TSV circuit models and the faults parameter extraction equations.

3. Multi-Tone Dither Test Method

3.1. The Principle of Multi-Tone Dither Test Method

Multi-tone dither test is a method of testing the circuit fault using the dynamic excitation signal. In this method, a synthetic multi-tone dither signal is used to stimulate the TSV circuit under test through the micro-strip lines. The PAR of the output response is measured by using a peak and envelope detector, which provides fault diagnosis [23]. The fault type and size are detected by comparing the degree of difference in PAR between the fault-free and faulty TSVs. In this method, the multi-tone dither signal is generated by using a multi-tone signal with additive white Gaussian noise to modulate an RF carrier signal of the same test frequency. The reason for choosing the multi-tone signal as the main excitation signal is that it does not suffer from spectrum spread or leakage, and more importantly, it is sensitive to phase distortion. It is important to test fault-free and faulty circuits under the same excitation conditions, as their output voltages have a phase difference. Since each tone undergoes a phase shift of $i\Delta\omega$ order during circuit testing (here, i is the number of tones), this cumulative phase shift variation results in a significant change in the V_{RMS} of the output signal of the circuit in which there is a fault [17, 18]. Therefore, this is a good mechanism for determining the defects in an electric circuit. Meanwhile, the phase shift varies depending on the type of faults, and its extent depends on the tone number of the multi-tone signal and the test frequencies. Furthermore, non-linear static errors in the PAR test result in harmonic distortions in the lost cell of the output spectrum. If a suitable dither noise is added, the out-

put noise spectrum will not overlap the spectrum of the test signal. Therefore, the additive white Gaussian noise is used to improve the sensitivity and robustness of the PAR test [24]. From the above, it can be seen that the multi-tone dither test effectively detects the manufacturing faults in TSVs.

The expression of the multi-tone signal is as follows:

$$u(t) = \sum_{i=0}^{N-1} |a_i| \cos(\omega_i t + \theta_i) \quad (10)$$

where a_i , ω_i , and θ_i are the amplitude, angular frequency, and initial phase of the i -th tone signal respectively, and i is the number of tones. The accuracy and sensitivity of the test method are changed by optimizing the design parameters to generate multi-tone signals [25].

The dither in the multi-tone dither test method is partly derived from the additive white Gaussian noise [24]. The edge of the multi-tone signal with the additive white Gaussian noise is no longer smooth and increases the amplitude information, which amplifies the PAR value and is useful for the fault test. Moreover, the multi-tone signal modulated with the RF carrier signal can compensate the attenuation of the signal in the transmission process and improves the fault coverage, which improve the accuracy of the entire TSV fault test [26].

The PAR is the ratio between the peak value of the voltage (V_{peak}) and V_{RMS} , which largely depends on the phase factor. In the multi-tone dither test, when there are faults in the TSV circuit, changes in impedance cause the changes in phase shift of the output waveform to become more obvious. This causes a significant change in the V_{RMS} of the output of the circuit. Therefore, in order to use the change in phase shift and its V_{RMS} effectively, the output response PAR of the circuit under test is represented as the test index. Its expression is as follows:

$$PAR = \frac{V_{peak}}{V_{RMS}} \quad (11)$$

where

$$V_{peak} = \max_t \{|u(t)|\} \quad (12)$$

$$V_{RMS} = \sqrt{\lim_{x \rightarrow N} \frac{1}{2x} \int_0^N u^2(t) dt} \quad (13)$$

3.2. Multi-Tone Signal Parameter Selection

As the selection of the parameters of the multi-tone signal is different, the resolution of the test results is also different [27, 28]. To improve the fault diagnosis rate, it is preferable to select the parameters of the multi-tone signal, including the tone number i , the fundamental angular frequency ω_0 , the change in angular frequency $\Delta\omega = 2\pi\Delta f_0$, the initial phase θ_i and amplitude a_i of each tone. On the one hand, it is evident that the higher the number of tones the better is the information retrieved from the circuit under test, and therefore the accuracy of the test method can be enhanced by increasing the number of tones. On the other hand, the higher the number of

Table 1. Differences in PAR of multi-tone dither signals at different test frequencies.

Number of tones	The signal frequencies		
	1 MHz	100 MHz	1 GHz
1	0.65	1.12	0.91
3	1.53	2.29	1.97
5	1.94	2.85	2.36
10	3.29	4.26	3.85
15	3.40	4.65	4.08

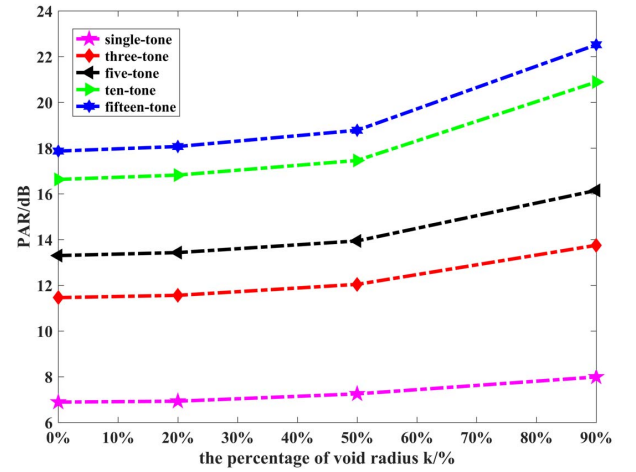
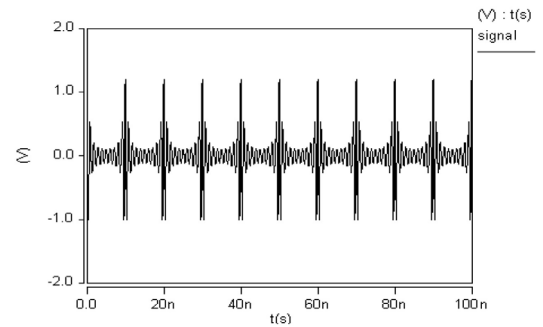
tones, the higher is the total RMS value of the generated signal. If this value is too high, problems related to circuit heating arise and cause complex signal generation. Consequently, to balance these factors, it is necessary to select a finite number of tones to generate a multi-tone signal. The amplitude range of each tone is determined by the working voltage and the number of tones. It is proved by experiments that setting the initial phase to 0 can make the PAR of multi-tone dither signals more different between fault-free and faulty TSV circuits. Therefore, regardless of the initial phase of the multi-tone signal, the first cosine signal is used as a reference, and the expression of the multi-tone signal used in the test is generated by cosine signal phase shifting as follows by Eq. (14).

$$u(t) = \sum_{i=0}^{N-1} |a_i| \cos(\omega_0 + i\Delta\omega)t \dots \dots \dots (14)$$

where a_i is the amplitude of the i -th tone, ω_0 is the fundamental angular frequency, and $\Delta\omega$ is the adjacent angular frequency interval.

Taking the void fault as an example, the void radius size corresponds to the missing proportion of the TSV copper conductor, which is used as the fault point for testing. The number of tones of the multi-tone signal is first selected to determine the value of N in Eq. (14). This is the only way in which the amplitude range of each cosine signal can be determined. Before selecting the number of tones, it is necessary to analyze the influence of multi-tone signals with different frequencies on the sensitivity of the multi-tone dither test. **Table 1** lists the difference between the PARs of the normal and the void GS-TSV for multi-tone dither signals with different frequencies and tones. The test results show that 100 MHz multi-tone dither signals have higher resolution for testing TSV faults.

For the following studies, some parameters are set as follows: the fundamental frequency $f_0 = 1$ GHz, $\Delta f = 0.1$ GHz, then the adjacent angular frequency interval is represented as $\Delta\omega = 2\pi\Delta f$. There is the influence of the tone number on the test resolution of the fault points to discuss. Single-tone and multi-tone signals with tones of 3, 5, 10, and 15 are added to the additive white Gaussian noise under same SNR conditions, and then they are modulated with the RF carrier signal. Afterwards, these synthetic signals are used to separately test the PAR of fault points corresponding to different void sizes. **Fig. 3** shows

**Fig. 3.** Multi-tone dither signals with different tone numbers for testing GS-TSV with void fault.**Fig. 4.** The 10-tone signal.

that as the number of tones increases, the resolution of the test excitation at each fault point is improved, and the size of the fault can be gradually distinguished. Although the resolution of the 15-tone test signal for voids is higher than that of the 10-tone signal, the effect is not obvious. Further, it may be noted that the larger the value of N , the greater is the difficulty in applying the test stimulus from the signal generator [29]. Accordingly, the 10-tone signal was selected as the optimized multi-tone signal for testing.

The working voltage of the GS-TSV with void equivalent circuit is 5 V and 10 tones are selected. Hence, the amplitude of each tone is not more than 0.5 V. Because of the multi-tone signal definition, the value of a_i can be randomly selected between 0 and 0.5 V. The 10-tone signal as shown in **Fig. 4** is created after the above series of steps.

3.3. TSVs with Fault Testing Process Based on Multi-Tone Dither Test

- **Step1:** Establish a TSV equivalent circuit model with manufacturing fault.

- **Step2:** Generate an optimized multi-tone signal on circuit simulation software platforms such as ADS and Saber, and then design and synthesize multi-tone dither test excitations.
- **Step3:** Test faultless TSV equivalent circuit. The multi-tone dither test excitation is applied to the faultless TSV equivalent circuit through micro-strip lines for simulation testing, and then the peak and envelope detector detects the output response (PAR).
- **Step4:** Test the faulty TSV equivalent circuit using **Step3**, change the types and sizes of manufacturing faults, then perform simulation tests and record the PARs. Finally, plot the relation chart between the PAR and the fault size.
- **Step5:** Generate the multi-tone dither test signal in **Step2** using the actual hardware circuit. First, generate the desired multi-tone signal by using an arbitrary waveform generator, and add additive white Gaussian noise to it. Secondly, use an RF signal generator to generate an RF carrier signal at the same frequency as the multi-tone signal, and modulate it with the multi-tone signal designed above to obtain a multi-tone dither test signal.
- **Step6:** The multi-tone dither test described in **Step3** is performed on the actual TSVs under test, and the PAR is recorded. Next, the actual measurement results are compared with the PARs in the relation chart in **Step4**, to judge whether there is a fault in the TSVs under test and what kind of fault exists.

4. TSV Fault Test Verification

Using the GS-TSV structure mentioned above, the test access is connected to the TSV circuit terminals to be tested through a micro-strip line with a length of $300\ \mu\text{m}$, and width and height of $1\ \mu\text{m}$. A GS-TSV with the following physical dimensions is assumed: $h_{TSV} = 100\ \mu\text{m}$, $r_{TSV} = 10\ \mu\text{m}$, $t_{ox} = 0.5\ \mu\text{m}$ and $P_{TSV} = 40\ \mu\text{m}$. The first step is to calculate the faultless GS-TSV electrical model parameters. The multi-tone signal used in the test is a 10-tone signal that is optimally designed by using the procedure described in Section 3.2. Furthermore, the frequency of the RF carrier signal used is 0.1 GHz and the amplitude is 250 mV.

The resulting synthesized multi-tone dither test signal is shown in **Fig. 5**.

4.1. Void Fault Simulation Test

Exploiting the previously synthesized multi-tone dither signal, the multi-tone dither method in Section 2 is used to test the GS-TSV with the void fault equivalent circuit. The location of the void also has some influence on fault detection. The closer the fault is to the top of the TSV,

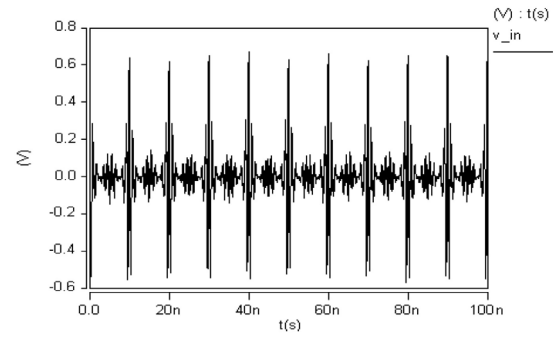


Fig. 5. Synthesized multi-tone dither test signal.

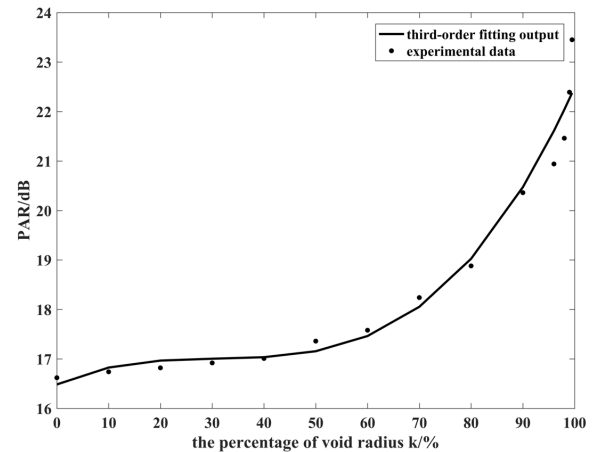


Fig. 6. The PAR of the void fault test.

where it is connected to the test access, the easier it is to be detected [11]. There is a void fault in the middle of the signal TSV. According to the analytical equations for voids in Section 2.2.1, the corresponding radius of the void (r_{void}) and fault resistance value (R_{open}) are calculated by changing the percentage k (ratio of the void radius to the TSV radius). The value of R_{open} is from $0\ \Omega$ (no fault) to $2.6k\ \Omega$ (strong resistive open). A simulation test, which mainly contains transient analysis, is conducted for each fault value, and measures the PAR of the output results. Performing the third-order fitting for the PARs of the measured data, the relationship between the PAR and the size of the void fault is shown in **Fig. 6**. The void radius increases with the value of k , and the total resistance increases, which causes a significant reduction in the V_{RMS} of the output signal. As expected, an increase in the resistance R_{open} leads to an increase in the PAR. In other words, the PAR of the equivalent fault circuit increases with the increase in void size. This indicates that resistive opens of sufficient size can be detected by measuring the PAR. As can be seen from **Fig. 6**, the PAR only begins to change gradually when the void fault's k is 10%, which corresponds to a void with a radius of $1\ \mu\text{m}$. Thus, void faults with a size as small as $1\ \mu\text{m}$ can be detected by the multi-tone dither test.

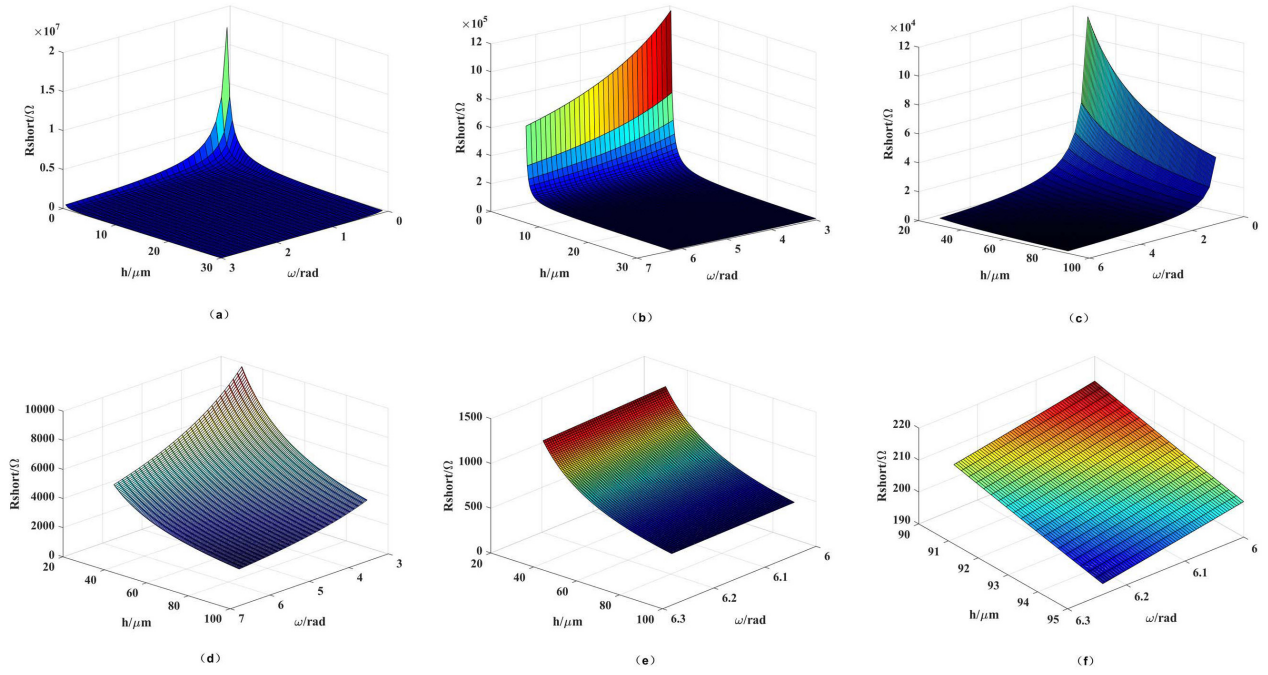


Fig. 7. Short impedance vs. pinhole fault size.

4.2. Pinhole Fault Simulation Test

The multi-tone dither signal is applied to the GS-TSV with the pinhole fault equivalent circuit for the simulation test. The impedance of the pinhole R_{short} as a two-variable function using the pinhole analytical Eq. (9) is depicted in Fig. 7. It obtains the corresponding fault value (R_{short}) based on the given pinhole height (h) and radian (ω). The multi-tone dither test is performed by selecting the impedance value of different pinhole sizes (h and ω). For each PAR value that is measured, the combination of all possible pinhole heights and radians are given by the intersection of the horizontal layer corresponding to the specific impedance value, with a 3D surface. The relationship between PAR and the impedance of the pinhole fault is obtained by the fifth-order fitting of PAR data, as shown in Fig. 8. As the impedance of the insulating layer decreases, more currents leak into the silicon substrate. The difference from the void is that the PAR of the equivalent circuit of the TSV pinhole fault decreases with the increase in pinhole size. As seen in Fig. 8, the approximate size of the pinhole impedance of the TSV can be determined by the PAR value. The multi-tone dither test could detect a weak pinhole fault impedance as small as $17.5 \text{ M}\Omega$, and the corresponding pinhole height and arc length are $0.5 \mu m$ and $1 \mu m$, respectively.

5. Conclusion

In this article, we choose the GS-TSV equivalent electrical model, and rigorously analyze typical manufacturing faults of TSV—void and pinhole faults, and then the

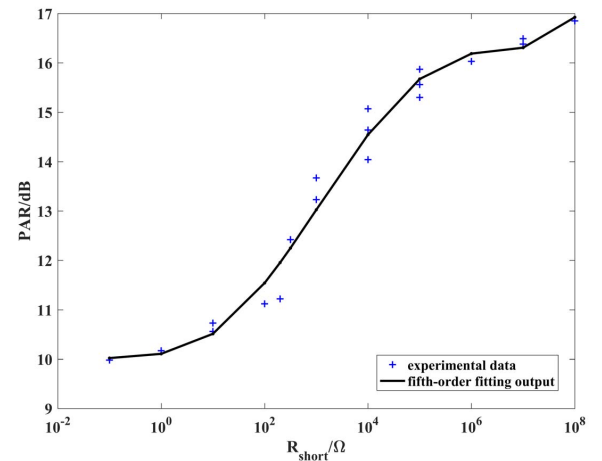


Fig. 8. The PAR of the pinhole fault test.

corresponding fault models are set up. A multi-tone dither test method is proposed for the equivalent circuit models of GS-TSV with voids and pinholes. This method mainly applies a modulated multi-tone signal with additive white Gaussian noise to stimulate the circuit under test, which can effectively solve the damage and problems caused by the probe test and the large-area peripheral test circuit design to the TSV, and diagnoses the TSV fault by analyzing the measured PAR. Simulation results using Saber and ADS CAD tools show that the feasibility of the proposal to detect a wide range of resistive faults of TSV, especially faults as small as several micrometers, can be effectively identified because of the significant differences in

PARs between faulty and fault-free TSV tests. Compared to some of the most representative test methods found in the literature, the present work has obvious advantages in fault detectability, area overhead and test time. It is able to detect lower critical open resistance and higher critical short resistance and determine the size of the manufacturing fault. For example, it is demonstrated that short impedances for pinholes up to several megohms are diagnosed and void open resistances of the order of ohms can be detected. Moreover, it is effective to detect different single manufacturing faults without any external test probe. In addition, the optimization of multi-tone signals can effectively improve the accuracy of the test. The results show that the test for weak faults is more robust at higher number of tones levels. This approach obviously enhances the sensitivity of weak fault tests and the accuracy of fault identification, providing new ideas and directions for TSV fault diagnosis. It is significant especially in dynamic excitation tests.

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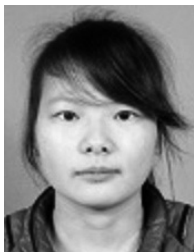
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